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Ultra-Stable Organic Electrochemical Transistors With Suppressed Swelling via Molecular Weight Modulation

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ABSTRACT

Organic electrochemical transistors (OECTs) are promising for next-generation bioelectronics due to their high performance and biocompatibility. Nevertheless, they still face tremendous operational stability challenges due to the limited robustness of the organic mixed ionic–electronic conductor (OMIEC) channel. Here, by modulating the molecular weight (MW) of OMIEC, enhanced OECT and relevant circuit operation stabilities are demonstrated, showing more than 3,000,000 full cycles (~42 h) with less than 15% current variation in an OECT, and 150,000 cycles (~4 h) with less than 5% voltage variation in an OECT-based inverter, which are among the highest of reported OECT-based electronics. Specifically, p(g2T-T), a typical p-type OMIEC, with varying MW (7–43 kDa), is synthesized, where lower-MW p(g2T-T) (~9 kDa) exhibits superior device performance and cycling stability in OECTs, outperforming those in high-MW counterparts (>30 kDa). It is indicated that low-MW p(g2T-T) maintains higher volumetric capacitance, ordered orientation, and reduced swelling. Therefore, irreversible microstructural degradation is effectively avoided, along with better performance yield. Furthermore, MW regulation enables physiological signal sensing with high tolerance to body fluid environments for 7 days. These findings highlight MW modulation as a versatile approach to suppress excessive swelling, advancing the design of durable OECT-based electronics.

1 | Introduction

Organic electrochemical transistors (OECTs) hold great potential in next-generation high-performance wearable bioelectronics and neuromorphic circuits, benefiting from the prominent advantages of low driving voltage [1–6], high transconductance (g_m) [7–13], and good biocompatibility [14–21]. However, long-term operation of OECTs and related electronics in aqueous electrolytes remains formidably challenging. Excessive swelling of the organic mixed ionic–electronic conductor (OMIEC)-based channel during successive ion doping/de-doping process can gradually alter and damage the OMIEC microstructure, leading to inevitable performance degradation [22, 23]. In addition, irreversible side reactions during transistor operation can also lead to OMIEC degradation and eventual failure of OECT-based electronics [24–26]. One of the key strategies to improve the operational stability of OECTs is to design OMIECs with enhanced durability [25, 27]. For instance, excessive swelling of OMIECs can be suppressed by rational redistribution of the oligo ethylene glycol (EG) side chains and the introduction of

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an alkyl spacer [28, 29]. Side reactions, such as the oxygen reduction reaction (ORR), can also be prevented by employing OMIECs with high ionization potential (≥4.9 eV) to eliminate the formation of H_2O_2 [30].

Molecular weight (MW), as an important parameter of conjugated polymers (CPs), has shown a significant effect on their optoelectronic and charge transport properties [31, 32]. For OMIECs, which are predominantly based on CPs, their MW also significantly affects the OECT performance and cycling stability [33, 34]. Fabiano et al. reported that in the ladder-type polymer BBL, the charge transport properties between the polymer chains were significantly improved by increasing the MW from 4.9 to 51 kDa [35]. Nguyen et al. reported that polyelectrolytes CPE-K with increased MW from 7.8 kDa (CPE-K_{12}) to 31.5 kDa (CPE-K_{48}) can effectively improve the charge percolation pathways, thereby improving OECT performances [36]. Rivnay et al. demonstrated the MW influence on the performance of glycolated polythiophene, P3MEEET [33], showing that OECT performance is improved from low MW to intermediate MW, then decreases in high MW. However, previous studies on MW regulation of OMIEC mainly focused on MW impact on the performance of OECTs, while no systematic research has been conducted on the effects of MW regulation on the ionic-electronic coupling, and especially, swelling and stability.

In this work, p-type p(g2T-T) [37–41] and n-type BBL OMIECs with different MWs are investigated in conventional planar OECTs (cOECTs) and vertical OECTs (vOECTs) in terms of their transistor performances and cycling stabilities. Specifically, a series of p(g2T-T) with different MWs (7–43 kDa) is synthesized. It is observed that with gradually increased MWs, inferior OECT performances are obtained, showing declined peak g_m and prolonged transient times (τ). Interestingly, the cycling stability of p(g2T-T) based OECTs also decreases with increasing MW, a trend that is similarly confirmed in BBL-based OECTs. As a result, OECT with p(g2T-T) MW of ∼9 kDa ($[g2T\text{-}T]_{15}$) exhibits the highest performance and cycling stability in both planar and vertical configurations, achieving ultra-high cycling stability of 3,000,000 full cycles. Further characterizations reveal a gradual decrease in volumetric capacitance (C^*) and a transition from ordered to disordered thin films with increasing MW. Electrochemical quartz crystal microbalance (EQCM) characterization reveals that high-MW p(g2T-T) undergoes significantly increased swelling during the doping/de-doping process. It is proposed that significantly increased swelling is mainly attributed to a higher proportion of amorphous regions. Consequently, increased swelling leads to significant separation between crystallites, resulting in breakdown of chain entanglement and film microstructure in high-MW p(g2T-T), causing a decrease in device performance and stability. Therefore, low-MW p(g2T-T) and BBL are utilized to construct vertically stacked complementary inverters (VSCIs) with ultra-high stability (more than 150,000 switching cycles with consistent voltage outputs). Furthermore, stable human electrooculogram (EOG) signal monitoring is demonstrated using VSCI with exceptional long-term stability in a body fluid environment. These results demonstrate the insights of MW influence on OECT performance and stability, providing a straightforward yet effective pathway for the construction of high-performance OECT-based electronics.

2 | Results and Discussion

p(g2T-T) was synthesized by Stille coupling polymerization following a previously reported synthetic route [39]. By changing the reaction time from 6 to 48 h, eight batches of p(g2T-T) with different MWs were successfully obtained, with number-average relative MWs (M_n) ranging from ∼7 to ∼43 kDa, they are referred to as $(g2T\text{-}T)_{12}$, $(g2T\text{-}T)_{15}$, $(g2T\text{-}T)_{30}$, $(g2T\text{-}T)_{37}$, $(g2T\text{-}T)_{40}$, $(g2T\text{-}T)_{46}$, $(g2T\text{-}T)_{64}$, and $(g2T\text{-}T)_{72}$, corresponding to the average number of repeating units of the polymers (Figure 1a). Note that MW of $(g2T\text{-}T)_{12}$, as determined by three parallel gel permeation chromatography (GPC) tests, yielded values of 51, 32, and 74 kDa, respectively, all significantly higher than the expected value. This discrepancy may stem from aggregate behavior in the low-MW polymers, preventing accurate MW determination by GPC. Therefore, the MW for the $(g2T\text{-}T)_{12}$ batch is derived from the linear fitting based on MW versus reaction time data from the other seven batches. (Figure S1). To comprehensively investigate the impact of p(g2T-T) with different MWs in OECTs, both cOECTs and vOECTs are fabricated (Figure 1b). Detailed fabrication procedures can be found in the Experimental Section. The channel geometries are 500 µm/20 µm/55 nm (channel *width[W]/length[L]/thickness[d]*) for cOECT, and 40 µm/55 nm/40 µm ($W/L/d$) for vOECT, respectively. Note, W, L, and d in vOECTs are defined by the bottom electrode width, OMIEC thickness, and top electrode width, respectively.

As shown in Figures 1c,d,g, S2 and Table S1, the transfer and g_m characteristics of cOECTs indicate that both on-current (I_{on}, drain current [I_D] under V_G of −0.5 V) and maximum g_m ($g_{m,max}$) decrease monotonously with the increasing MW (except those from $[g2T\text{-}T]_{12}$, which holds the lowest MW). Specifically, at V_D of −0.1 V, I_{on} of −3.21 ± 0.18 mA in $(g2T\text{-}T)_{15}$ gradually decreases to −0.36 ± 0.02 mA in $(g2T\text{-}T)_{72}$, while $g_{m,max}$ decreases by more than 81% (from 11.40 ± 0.45 mS in $[g2T\text{-}T]_{15}$ to 2.20 ± 0.15 mS in $[g2T\text{-}T]_{72}$). For vOECTs, significantly enhanced transistor performances are obtained when compared to those in cOECTs, where approximately one order of magnitude higher I_{on} and $g_{m,max}$ are achieved (Figures 1e,f,h, and S3), and all vOECTs fabricated from eight batches of p(g2T-T) with varying MWs exhibit high device yield (>87.5%). Only a small number of devices near the substrate edges may fail to function properly. Meanwhile, both I_{on} and $g_{m,max}$ of vOECTs still show a decreasing trend when MW grows from $(g2T\text{-}T)_{15}$ to $(g2T\text{-}T)_{72}$ (Table S2). Similarly, at $V_D = -0.1$ V, I_{on} and $g_{m,max}$ decrease from −14.57 ± 0.97 mA and 99.18 ± 6.32 mS in $(g2T\text{-}T)_{15}$ to −11.32 ± 0.55 mA and 71.13 ± 2.14 mS in $(g2T\text{-}T)_{72}$, respectively. In addition, monotonous decay trends in device performance can also be observed in both cOECTs and vOECTs under $V_D = -0.5$ V. Note that, compared with cOECTs, the decay trend in performance of p(g2T-T) with different MWs is less pronounced in vOECTs. This is due to the low channel resistance of vOECTs under ON states, which makes the total resistance of the vOECT mainly determined by the contact and series electrode resistances (Figures S4 and S5).

Next, transient responses of both cOECTs and vOECTs based on p(g2T-T) are investigated (Figures S6–S9). The turn-on transient time (τ_{ON}) and turn-off transient time (τ_{OFF}) are extracted and summarized in Figure 1i,j, Tables S1 and S2. For cOECTs, $(g2T\text{-}T)_{12}$ shows the fastest τ_{ON} of 43.21 ± 2.47 ms and τ_{OFF} of

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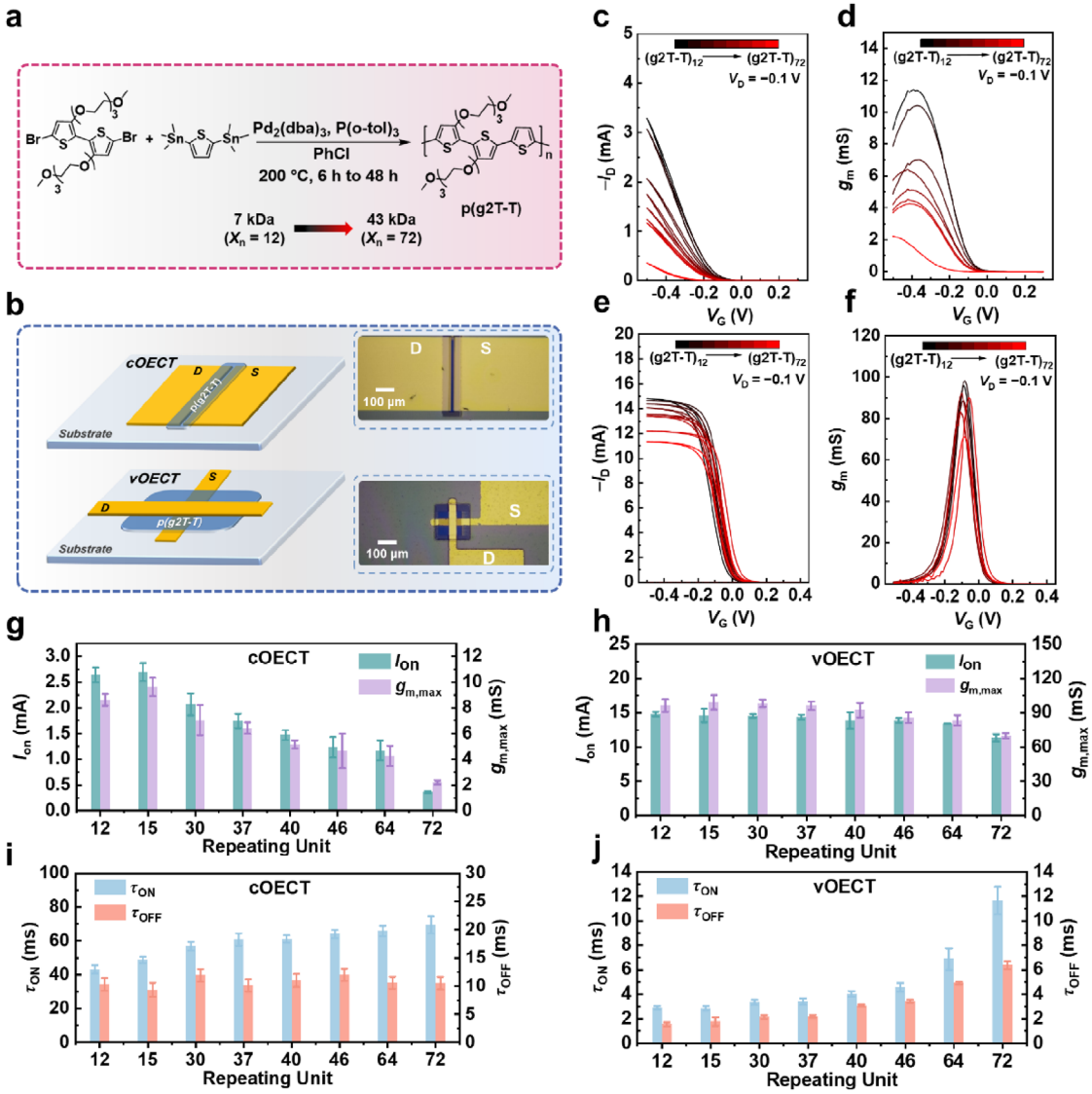


FIGURE 1 | (a) Synthetic route of p(g2T-T) with different MWs. (b) Device schematics and optical images of cOECT and vOECT. (c) Transfer and (d) g_m characteristics of cOECTs with different MWs of p(g2T-T). (e) Transfer and (f) g_m characteristics of vOECTs with different MWs of p(g2T-T). I_{on} and $g_{m,max}$ of (g) cOECTs and (h) vOECTs based on different p(g2T-T) MWs. τ_{ON} and τ_{OFF} of (i) cOECTs and (j) vOECTs based on different p(g2T-T) MWs ($V_D = -0.1$ V).

10.29 ± 1.11 ms. As MW gradually increases, τ_{ON} and τ_{OFF} increase to 69.36 ± 5.12 and 10.51 ± 1.11 ms, respectively. The same trend is observed in vOECTs, with overall shorter τ (Figures S8 and S9). Note that τ reported here is not as fast as that in the previous report, which is influenced by the fabrication process of the SU-8 encapsulation layer, where it may introduce thermal annealing of the channel [42]. Moreover, at a high V_D of –0.5 V, slower switching of both cOECTs and vOECTs is observed (Figure S10), which is due to the weaker electric field generated between the drain and gate electrodes, resulting in slower ion diffusion.

The above results demonstrate that the performances of both cOECTs and vOECTs exhibit a decreasing trend with the increasing MW of p(g2T-T), significantly different from previously reported OMIECs (such as BBL and CPE-K) [35, 36]. This suggests that, for OMIECs with oligo EG side chains, such as p(g2T-T), the increased carrier mobility due to higher MW is not the decisive factor influencing their performance in OECTs.

Compared to previously reported patterns in P3MEEET [33], when operating in NaCl, except for P3MEEET-3, both low- and high-MW P3MEEET exhibit negligible mobility differences due to their comparable crystallinity. However, changes in volume capacitance due to potential variations in film morphology result in higher μC^* for high-MW P3MEEET compared to their low-MW counterparts, thereby enabling better device performance. The contradictory MW-performance relation observed between P3MEEET and p(g2T-T) indicates that for glycolated OMIECs, the influence of MW on OECT is anticipated to be material dependent. The introduction of polar side chains exerts dual effects on both film crystallinity and swelling deformation. This dual influence renders the relation between MW and device performance more complex, varying significantly across different

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OMIECs. Therefore, this work does not conduct a comprehensive investigation into the universal relation between MW and OECT performance. Instead, it primarily focuses on studying the impact of MW regulation on the cycling stability of OECTs.

Next, OECTs with the highest and lowest device performance ($[g2T\text{-}T]_{15}$ and $[g2T\text{-}T]_{72}$), as well as those with two intermediate MWs ($[g2T\text{-}T]_{30}$ and $[g2T\text{-}T]_{46}$), were selected for further study. To conduct a more comprehensive investigation of the impact of MW regulation on cycling stability, we performed cycling stability tests with OECTs operating in linear ($V_D = -0.1$ V) and saturation regimes ($V_D = -0.5$ V), respectively. Due to differences in transient responses (for cOECTs, $\tau_{ON} > 40$ ms; for vOECTs, $\tau_{ON} < 10$ ms) across device structures, different cycling frequencies are set to ensure that OECTs can fully turn ON or OFF during each cycle. In the meantime, to maintain a fixed total test duration for cOECT and vOECT (10,000 s) for a fair comparison, cOECT and vOECT underwent a total of 20,000 and 200,000 cycles, respectively. Cycling stability measurement of cOECTs is first conducted by continuously monitoring I_D change over time upon applying a square wave of V_G (Figures S11 and S13), where a transfer characteristic is conducted after every 2000 switching cycles (Figures S12 and S14). When operating in the linear regime, cOECT based on $(g2T\text{-}T)_{15}$ exhibits the best cycling stability with 52.60% and 53.43% of initial I_D and $g_{m,max}$ retained, respectively, after 20,000 cycles (Figure 2a,b). Meanwhile, significant degradation in $(g2T\text{-}T)_{72}$-based cOECT is observed with only 14.70% and 9.23% of the initial I_D and $g_{m,max}$ retained, respectively. When operating in the saturation regime, the cycling stability of all cOECTs shows a significant decline, primarily due to the high gate bias combined with non-uniform charge distribution in the saturation regime (Figure 2c,d). Nevertheless, $(g2T\text{-}T)_{15}$-based cOECT still exhibits the best cycling stability, showing 44.86% and 45.27% of initial I_D and $g_{m,max}$, respectively. While $(g2T\text{-}T)_{72}$-based cOECT shows only 11.59% and 8.33% of the initial I_D and $g_{m,max}$ after 8000 cycles. Diverse behaviors of p(g2T-T) with different MWs in cOECTs suggest that increased MW not only weakens the device performance but also jeopardizes the cycling stability.

Meanwhile, in vOECTs (Figures S15 and S16), tremendously different cycling stabilities in the linear regime are observed. $(g2T\text{-}T)_{15}$-based vOECT exhibits good cycling stability with 63.13% and 52.90% of initial I_D and $g_{m,max}$ retained, respectively, after 200,000 cycles. Meanwhile, $(g2T\text{-}T)_{30}$ and $(g2T\text{-}T)_{46}$ show gradually decreasing cycling stability. Particularly, I_D of $(g2T\text{-}T)_{72}$-based vOECT undergoes significant fluctuations during the first 20,000 cycles, followed by a device failure after 45,000 cycles (Figure 2e,f). This indicates that $(g2T\text{-}T)_{72}$ may have greater swelling during cycling (vide infra), resulting in film rupturing and detachment, causing a short circuit between the top and bottom electrodes [43]. Note, tremendously enhanced stability in vOECTs is assigned to the vertical structure, where the top electrode would prohibit the swelling [41]. When operating in the saturation region, the cycling stability of most vOECTs (except $[g2T\text{-}T]_{72}$) is further enhanced (Figures S17 and S18). Specifically, $(g2T\text{-}T)_{15}$ and $(g2T\text{-}T)_{30}$ both maintain over 95% of initial I_{on} and $g_{m,max}$, while $(g2T\text{-}T)_{46}$ also maintains 86.20% and 77.69% of initial I_{on} and $g_{m,max}$, respectively (Figure 2g,h). This observation contrasts with the cycling stability in cOECT. We attribute these results primarily to the significantly reduced channel length in vOECTs. At higher V_D, the Coulombic interaction between source and drain electrodes in the vertical direction is intensified, which further suppresses channel swelling and consequently improves device cycling stability. For $(g2T\text{-}T)_{72}$, excessive swelling may have exceeded the threshold for Coulombic interactions, causing the OMIEC to lose the protective effect of the top electrode, leading to a rapid performance degradation. These results underscore the advantages of utilizing vOECTs as fundamental building blocks for achieving highly stable logic circuits, biosensors, and bioelectronic devices.

Besides, we have also conducted additional cycling stability of vOECTs under the same biasing conditions as cOECTs (Figure S19). At this lower gate pulse frequency, the cycling stability of the $(g2T\text{-}T)_{15}$-based vOECT shows significant improvement, while the $(g2T\text{-}T)_{72}$-based vOECT still exhibits poor cycling stability. Both vOECTs still demonstrate enhanced cycling stability when compared to cOECTs. It is observed that reduced total cycling number and lowered frequency stress will relieve the channel deterioration, but they still cannot compensate for the stability differences caused by variations in material swelling.

Thus, OECTs based on high-MW p(g2T-T) all demonstrate inferior cycling stability, indicating high MW in this case is not as expected to be more stable than those in the case of organic field effect transistors [44, 45].

In addition, cycling stability of $(g2T\text{-}T)_{12}$-based OECTs is also conducted to verify whether a lower MW can further improve stability (Figures S20 and S21). It is shown that $(g2T\text{-}T)_{12}$ is less stable than $(g2T\text{-}T)_{15}$ in both device configurations (Figure 2). This may be due to increased crystallinity and reduced film continuity at lower MW, leading to increased film defects and accelerating material deterioration during cycling [45].

Furthermore, since the vOECT based on $(g2T\text{-}T)_{15}$ shows the best cycling stability with no obvious decay up to 10,000 cycles, an extremely long cycling stability evaluation is conducted under lower cycling V_G bias (+0.3 to −0.4 V) (Figures 2i,j, and S22) to improve cycling stability [40]. With V_D of −0.1 V, such OECT can continuously operate for 3,000,000 full cycles (~42 h), with 86.22% of initial I_D and 51.45% of initial $g_{m,max}$ retained, far exceeding all the reported stability results of OECTs [2, 8, 10, 21, 24, 25, 28, 30, 40–42, 46–49] (Figure 2k, Table S3). An additional long-term stability evaluation with identical testing bias is further conducted, which further verifies the good reproducibility of the cycling stability (Figure S23). With a V_D of −0.5 V, such an OECT can still operate for 3,000,000 full cycles with 59.73% of initial I_D and 63.58% of initial $g_{m,max}$ retained (Figure S24). Whether operating in the linear regime or the saturation regime, $(g2T\text{-}T)_{15}$-based vOECT demonstrates the longest continuous operating time and highest cycling number, fully demonstrating the exceptional stability of this OECT.

Enhanced stability by means of low MW of OMIEC is further verified by employing BBL with two different MWs (BBL_{45} [15 kDa] and BBL_{105} [35 kDa]) as the channel of OECTs. Transfer, output, and transient characteristics of BBL-based OECTs can be found in Figures S25–S27. Note that the current overshoot observed in the transient curves of BBL-based cOECT in Figure S27 is mainly induced by the capacitive charging/discharging process of the channel capacitance; this phenomenon can be

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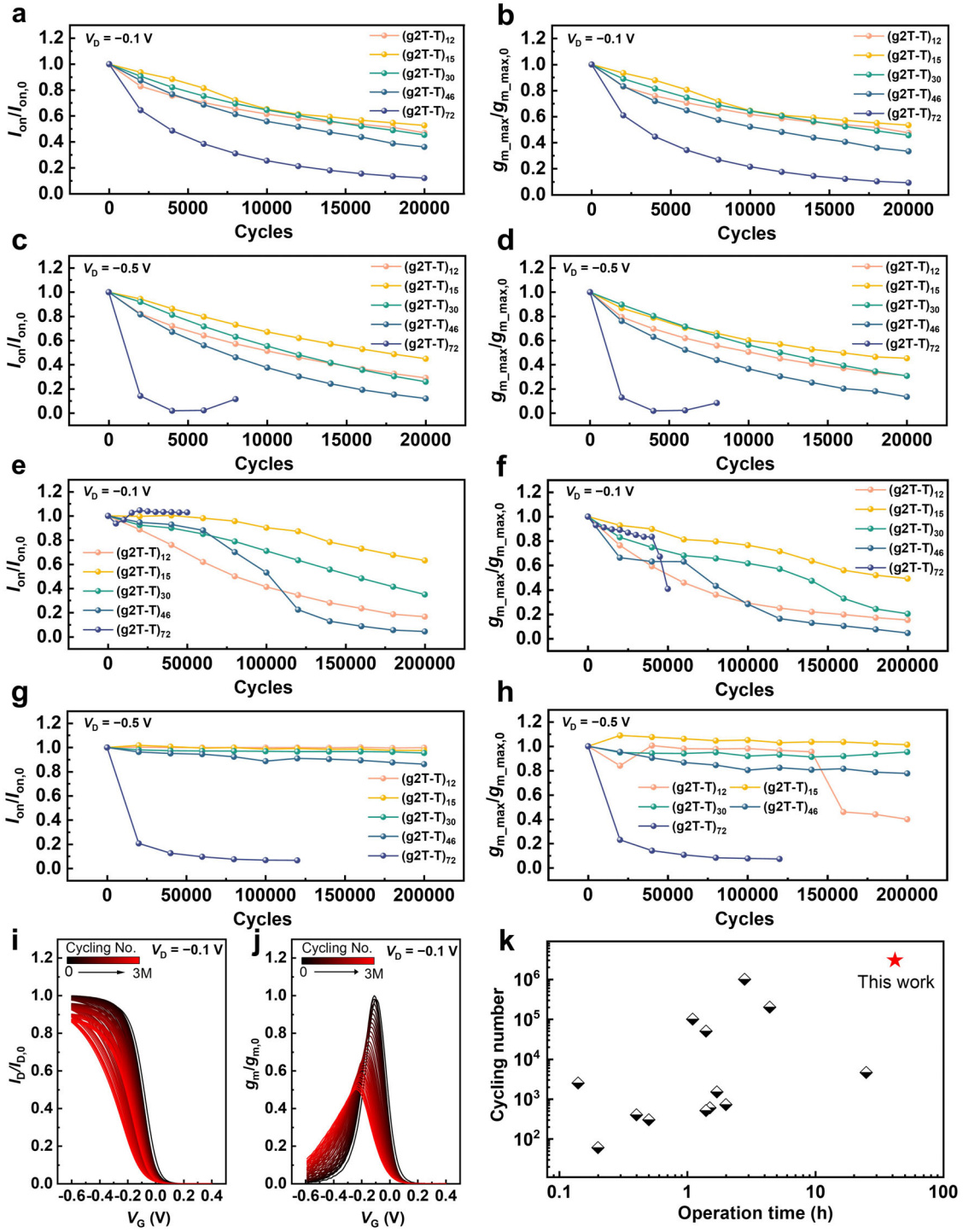


FIGURE 2 | Cycling stability of OECTs. (a, c) I_{on} and (b, d) g_m decay trends of $(g2T\text{-}T)_{12}$, $(g2T\text{-}T)_{15}$, $(g2T\text{-}T)_{30}$, $(g2T\text{-}T)_{46}$, and $(g2T\text{-}T)_{72}$ in cOECTs under different V_D. (e, g) I_{on} and (f, h) g_m decay trends of $(g2T\text{-}T)_{12}$, $(g2T\text{-}T)_{15}$, $(g2T\text{-}T)_{30}$, $(g2T\text{-}T)_{46}$, and $(g2T\text{-}T)_{72}$ in vOECTs under different V_D. Decay trends of (i) transfer and (j) g_m characteristics of $(g2T\text{-}T)_{15}$-based vOECT during 3,000,000 full cycles. (k) Comparison of cycling number and operation time values with other OECTs with high cycling stability reported in the literature.

clearly observed when the channel current is low, and it has also been reported in previous literature [35, 50]. Cycling stability measurement of BBL-based OECTs is also conducted with a biasing frequency of 10 Hz (Figures S28 and S29). cOECT based on BBL_{45} exhibits better cycling stability than that with BBL_{105}. In vOECTs, both BBL_{45} and BBL_{105} exhibit comparable degradation in cycling stability (Figure S30). These cycling behaviors observed in both cOECTs and vOECTs are expected to correlate with the MW differences.

To get further insights into the film properties with different MWs, the physicochemical properties of those OMIEC thin films are investigated by UV-vis-NIR spectra. Steady-state UV-vis-NIR spectra of p(g2T-T) films with different MWs show that all films

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exhibit two absorption peaks located in the range of 610–670 nm and can be ascribed to 0–1 and 0–0 vibrational peaks (Figure S31a). The locations of 0–0 vibrational peak exhibit randomly shifting along with irregular variations of 0–1/0–0 ratios, suggesting that the conjugation degree and π–π stacking of p(g2T-T) films with MW > 10 kDa show no significant trend with increasing MW (Figure S31b). Spectroelectrochemistry measurements are then performed to investigate the optical band variation upon electrochemical doping (Figures S32–S34). It is indicated that all p(g2T-T) films exhibit apparent ground state bleaching located at 400–700 nm, accompanied by the formation of a polaron with more than 0.3 a.u. absorption intensity increasing occurring at 700–1200 nm, suggesting that negative bias can efficiently drive anions into the bulk of p(g2T-T) films, inducing the formation and accumulation of hole carriers on the backbone of p(g2T-T). In addition, by extracting absorbance variations at fixed wavelengths (Figure S34), all films exhibit formation of polaronic species beyond a voltage of 0.2 V, indicating MW differences hardly affect the ion injection and polaron formation threshold of p(g2T-T).

Next, to validate the effect of variations in film microstructure with increasing MW on the C^* of OECTs, electrochemical impedance spectroscopy (EIS) characterization is further conducted to extract the value of C^*. As shown in Figures 3a and S35–S39, it is observed that C^* continuously decreases from 299.80 to 165.52 F cm^{-3} with increasing MW. It seems that increasing MW of p(g2T-T) may result in the transition of microstructures from higher electrochemical activity to lower electrochemical activity; thus, the ionic doping efficiency is reduced. This result is probably due to a decrease in electrochemically active film fraction [51].

Then, grazing-incidence wide-angle X-ray scattering (GIWAXS) was employed to investigate microstructural variations of p(g2T-T) thin films with different MWs (Figures 3b,c and S40). Pronounced (100) scattering can be observed out-of-plane for all polymers, suggesting edge-on-dominant orientation, which is consistent with previous reports [35–37]. Herman's Orientational Factors (f_x) [52] were extracted from the azimuthal plot (Figure S40b) for quantitative analysis of the crystal anisotropy, where similar values around 0.6 were found for all four polymers, suggesting a highly textured edge-on preference (Table S4). The q value for the (100) scattering is found at 4.19 nm^{-1}, which is the lowest among all four polymers, corresponding to the largest lamellar stacking distance (Figure S40c). This variation is consistent with what has been reported in previous literature, indicating that higher structural ordering at lower MW dominates the charge transport pathways [38]. A drastic decrease in scattering intensities is observed for the (100) scatterings with the increase of MW, suggesting a gradual decline in crystallinity, indicating the increasing proportions of amorphous region in the high MW polymers (Figure 3b,c). Interestingly, the crystal coherence length (CCL) extracted from the full width at half maximum (FWHM) of the (100) scatterings using the Scherrer Equation indicates almost identical crystallite sizes (~15 nm) in all polymers, despite the significant variations in crystallinity (Figure S40d, Table S4). This is likely due to the polymers being extended in low-MW polymer crystals, while the crystalline high-MW polymers are folded or extended into the amorphous region [53]. As the crystalline domains provide significantly higher charge transport abilities, higher crystallinity in low-MW polymers accounts for the increased OECT performances. Atomic force microscope (AFM) characterization on p(g2T-T) films with different MWs are conducted (Figure S41), the surface roughness (σ_{RMS}) ranges from 1.1 nm for $(g2T\text{-}T)_{15}$ to 1.5 nm for $(g2T\text{-}T)_{46}$ with polycrystalline-like morphology, while $(g2T\text{-}T)_{72}$ exhibits increased surface roughness of 2.1 nm with irregular granular particles, indicating a decrease in crystallinity, consistent with GIWAXS results.

Therefore, electrochemical quartz microbalance (EQCM) is employed to investigate the swelling behavior of p(g2T-T) films (Figure 3d,e) [51, 54]. The relative mass changes of p(g2T-T) films under biasing are quantified to evaluate the swelling differences of p(g2T-T) with different MWs. It is observed that with the gradual increase in MW, the relative mass change due to passive swelling (film swelling after exposure to electrolyte without applying bias) increases significantly, from 5.8% ($[g2T\text{-}T]_{15}$) to 20.6% ($[g2T\text{-}T]_{72}$). Such a phenomenon indicates that, in high MW p(g2T-T), ingress of electrolyte leads to more pronounced disorder in polymer microstructures [51]. When a bias of 0.5 V (vs. Ag/AgCl) is applied, all films experience a further increase in mass (film swelling in electrolyte with applied bias, referred to as active swelling), indicating efficient anionic doping. Among them, $(g2T\text{-}T)_{15}$ shows the minimal active mass increase (8.3%) during doping, which is conducive to enhancing its cycling stability in aqueous electrolytes. As MW increases, the active doping mass change increases further, with $(g2T\text{-}T)_{72}$ exhibiting the largest change of 31.8%. After the doping process is completed and the bias is reduced to 0 V, all films can be effectively de-doped. When a bias of −0.3 V (vs. Ag/AgCl) is applied, a slight increase in mass is observed in all four films, mainly attributed to the partial injection of hydrated cations from the electrolyte.

During device operation, excessive swelling of p(g2T-T) film may result in separation of crystallites and disruption of percolation, thus hampering the OECT performance [33]. Meanwhile, extensive swelling can also accelerate the deterioration and degradation of OECTs, consistent with our cycling stability results [55]. To quantitatively analyze the dominant effect on cycling stability, scatter diagrams with the $g_{m,max}$ retention rate of cOECT and vOECT against the changes in C^*, crystallinity, and swelling mass for correlation analysis are plotted in Figures 3f,g and S42. These results indicate that in both OECT structures, swelling and cycling stability exhibit the highest Pearson correlation coefficients ($|r| > 0.95$), suggesting that swelling exerts the most significant influence on cycling stability. C^* and scattering intensity of the films also demonstrate a relatively high correlation with cycling stability. Note that scattering intensity is obtained by integration of the (100) scattering from the out-of-plane linecut profile. Though scattering intensity in arbitrary units is directly correlated with the crystallinity, without the knowledge of the scattering intensity of 100% crystal obtained with an identical setup, an absolute crystallinity value is unable to be calculated here; therefore, only a semi-quantitative analysis of crystallinity is conducted here.

Therefore, it is proposed that the increased mass change in active swelling of p(g2T-T) films with increasing MW is the primary cause of the decline in cycling stability. For CPs, MW significantly influences their nanostructure, which causes distinct contrast in nanostructure development [53]. During the crystallization of a well-studied polymer semiconductor, P3HT, which shares

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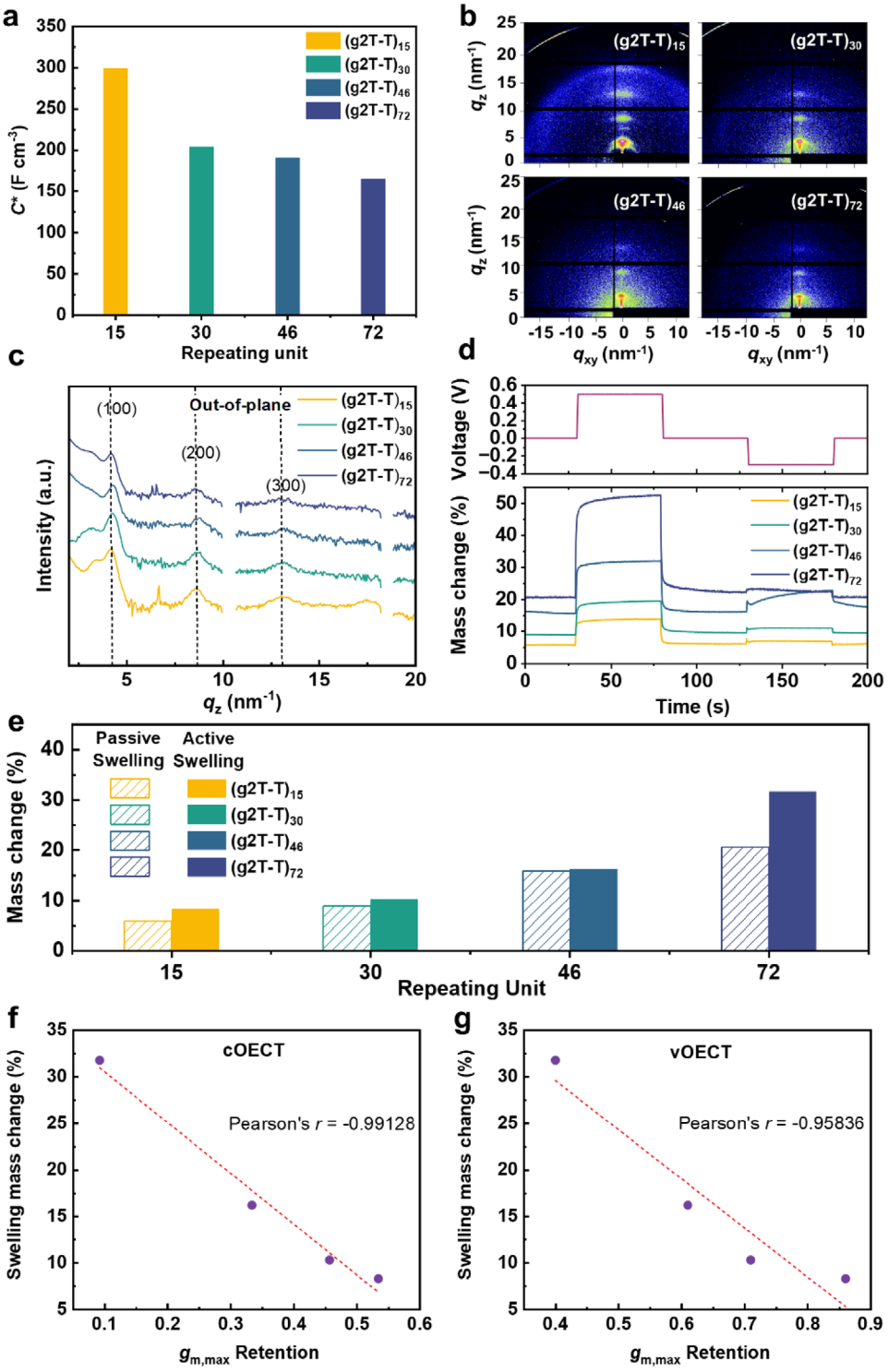


FIGURE 3 | (a) Volumetric capacitance calculated from EIS. (b) 2D GIWAXS patterns of $(g2T\text{-}T)_{15}$, $(g2T\text{-}T)_{30}$, $(g2T\text{-}T)_{46}$, and $(g2T\text{-}T)_{72}$ thin films. (c) Out-of-plane GIWAXS linecuts of p(g2T-T) thin films. (d) Relative mass change in response to square-wave potential sweeping, obtained from EQCM, along with the (e) active and passive mass change of p(g2T-T) thin films. Fitting results of correlation analysis between the $g_{m,max}$ retention rate of (f) cOECTs and (g) vOECTs, and swelling mass change.

an identical backbone with p(g2T-T), a non-folded stacking lamellar system is found with low-MW polymers (M_n < 20 kDa), whereas the higher-MW counterparts form folded crystallites interconnected with so-called "tie molecules" dangling through the amorphous region [53, 56]. Due to the lack of distinct amorphous regions resulting from lower chain entanglement, low-MW P3HT tends to readily form ordered arrangements during film formation, thereby exhibiting higher crystallinity [53]. This crystallization process also applies to low-MW p(g2T-T). However, high-MW P3HT exhibits notably higher charge transport mobilities due to effective charge transport between crystallites through the tie chains [53, 57, 58]. On the other hand,

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the tie molecules in high-MW polymers increase the viscosity of cooled solutions significantly [59] and are found to elongate during mechanical deformation, either during drawing [53, 60], compression [61], or friction [62]. The elongated tie molecules provide high-MW polymers with outstanding mechanical as well as electronic performances [53, 60].

On the other hand, for polymers with glycolated side chains such as p(g2T-T), previous reports have elucidated that anion uptaking induces lamellar expansion and water accumulation in amorphous regions for mass modulation [28, 43, 51, 63]. Swelling primarily occurs in the amorphous regions of p(g2T-T), as low-MW p(g2T-T) holds higher crystallinity, demonstrating a smaller portion of amorphous regions for swelling, enabling effective ionic doping of the conjugated backbone and resulting in a higher C^* value. On the contrary, although tie molecules serve as a key factor in enhancing charge transport and mechanical properties in high-MW P3HT, in high-MW p(g2T-T) with hydrophilic side chains, the substantial proportion of amorphous regions allows considerable water accumulation. This not only causes lamellar expansion but also leads to significant volume expansion, which may substantially disrupt the bridging effect of tie molecules. Consequently, continuous operation in aqueous electrolyte gradually generates significant separation between crystallites, impairs effective ionic doping of the conjugated backbone, causing a reduction in C^*. These can lead to the breakdown of chain entanglement and film microstructure in high-MW p(g2T-T), and finally, unstable OECT operation [32, 33, 45, 53, 64].

In addition, since prior literature has reported that higher residual palladium catalyst content in high-MW polymers reduces OECT performance and stability [65], we performed a quantitative analysis of residual Pd concentration in three p(g2T-T) batches with distinct MWs ($[g2T\text{-}T]_{15}$, $[g2T\text{-}T]_{30}$, and $[g2T\text{-}T]_{72}$) using inductively coupled plasma-mass spectrometry (ICP-MS). The results indicate that the Pd content in $(g2T\text{-}T)_{15}$, $(g2T\text{-}T)_{30}$, and $(g2T\text{-}T)_{72}$ is 751.83, 550.20, and 647.06 ppb, respectively. Since no clear pattern exists in Pd concentration across different MWs of p(g2T-T), we conclude that the primary factor affecting cycling stability with MW variation does not stem from residual Pd catalyst in the polymer, but rather from varying degrees of swelling of the films.

Next, VSCIs are fabricated using low-MW OMIECs, where $(g2T\text{-}T)_{15}$:DtFDA and BBL_{45} serve as p-type and n-type channel materials, respectively (Figure 4a) [39]. Note that there are two main purposes for adding photo-crosslinker, DtFDA, in p-channel: first, to prevent the top layer from delamination due to swelling of the underlying BBL; second, to change the laser direct writing patterning to photo-patterning, thereby avoiding damage to the bottom BBL from the laser. Rail-to-rail voltage transfer curves with switching voltages (V_{SW}) centered at near $V_{DD}/2$ (V_{SW} are 0.06, 0.14, 0.17, 0.24, 0.29, and 0.34 V for V_{DD} of 0.1, 0.2, 0.3, 0.4, 0.5, and 0.6 V, respectively) are obtained (Figure 4b). Meanwhile, when VSCI operates at V_{DD} as low as 0.1 V, a voltage gain of 2.4 V/V and a low power consumption of 6.02 μW are obtained. When V_{DD} = 0.6 V, voltage gain reaches 27.5 V/V, with power consumption of 1.54 mW (Figures 4b and S43). During practical testing, we also observed a significant improvement in VSCI stability at reduced driving voltages due to improved stability under lower biasing [40]. Consequently, to demonstrate longer cycling stability, we selected a lower drive voltage of V_{DD} = 0.3 V, and the VSCI can operate continuously for over 150,000 switching cycles (Figure 4c) due to the excellent stability of p- and n-channel materials. Note that the bumps in the cycling curve of Figure 4c from HIGH to LOW primarily stem from the BBL_{45}-based vOECT unit not yet reaching full steady-state operation during the initial phase of cycling testing. As the cycling testing continues, the τ_{ON} of BBL_{45}-based vOECT decreases (Figure S44), and such bumps gradually diminish. Leveraging the high stability of this VSCI, real-time monitoring of human EOG is then conducted (Figure 4d). Amplified EOG signals up to 10 mV are obtained by VSCI in monitoring horizontal eye movement. In addition, the VSCI demonstrates excellent storage stability, maintaining good amplification capability for EOG signal monitoring (signal amplitude remained at ~10 mV) after being immersed in PBS buffer for seven days (Figure 4e). The high stability and tolerance to body fluid environments of the VSCI highlight its potential for further applications in the field of bioelectronics.

3 | Conclusion

In this study, we systematically investigate the effect of MW modulation on the cycling stability of OECTs. For the p-type OMIEC p(g2T-T), low MW (~9 kDa) exhibited outstanding device performance and cycling stability in both planar and vertical architectures, outperforming their high-MW counterparts. Furthermore, vOECT based on $(g2T\text{-}T)_{15}$ achieved unprecedented ultra-high stability of up to 3,000,000 full cycles. Electrochemical characterization and structural analyses revealed that low-MW polymers exhibited higher C^*, as well as increased crystallinity, facilitating efficient charge transport, thereby improving OECT performance. EQCM studies further demonstrated that low-MW polymers minimized both passive and active swelling during electrochemical doping, mitigating irreversible microstructural breakdown and material degradation. Remarkably, this MW-dependent stability trend was validated in an n-type OMIEC–BBL, where the lower-MW variant (BBL_{45}) shows enhanced cycling stability, underscoring the universality of this approach. Moreover, the integration of $(g2T\text{-}T)_{15}$ and BBL_{45} enabled the fabrication of VSCI with high stability and endurance in human EOG monitoring, highlighting the practical viability of MW modulation for advanced bioelectronics. These findings establish MW regulation as a versatile strategy to optimize ionic-electronic coupling while suppressing detrimental swelling, thereby addressing a critical bottleneck in OECT durability. This work not only advances the fundamental understanding of structure-property relations in OMIECs but also provides a scalable pathway for designing robust, long-lasting OECTs for next-generation biomedical and neuromorphic applications.

4 | Experimental Section

4.1 | Materials

All reagents and chemicals were purchased from commercial sources and were used without further purification. 5,5′-Dibromo-3,3′-bis(2-(2-(2-methoxyethoxy)ethoxy)ethoxy)-2,2′-bithiophene were purchased from SunaTech, 2,5-bis(trimethyl-

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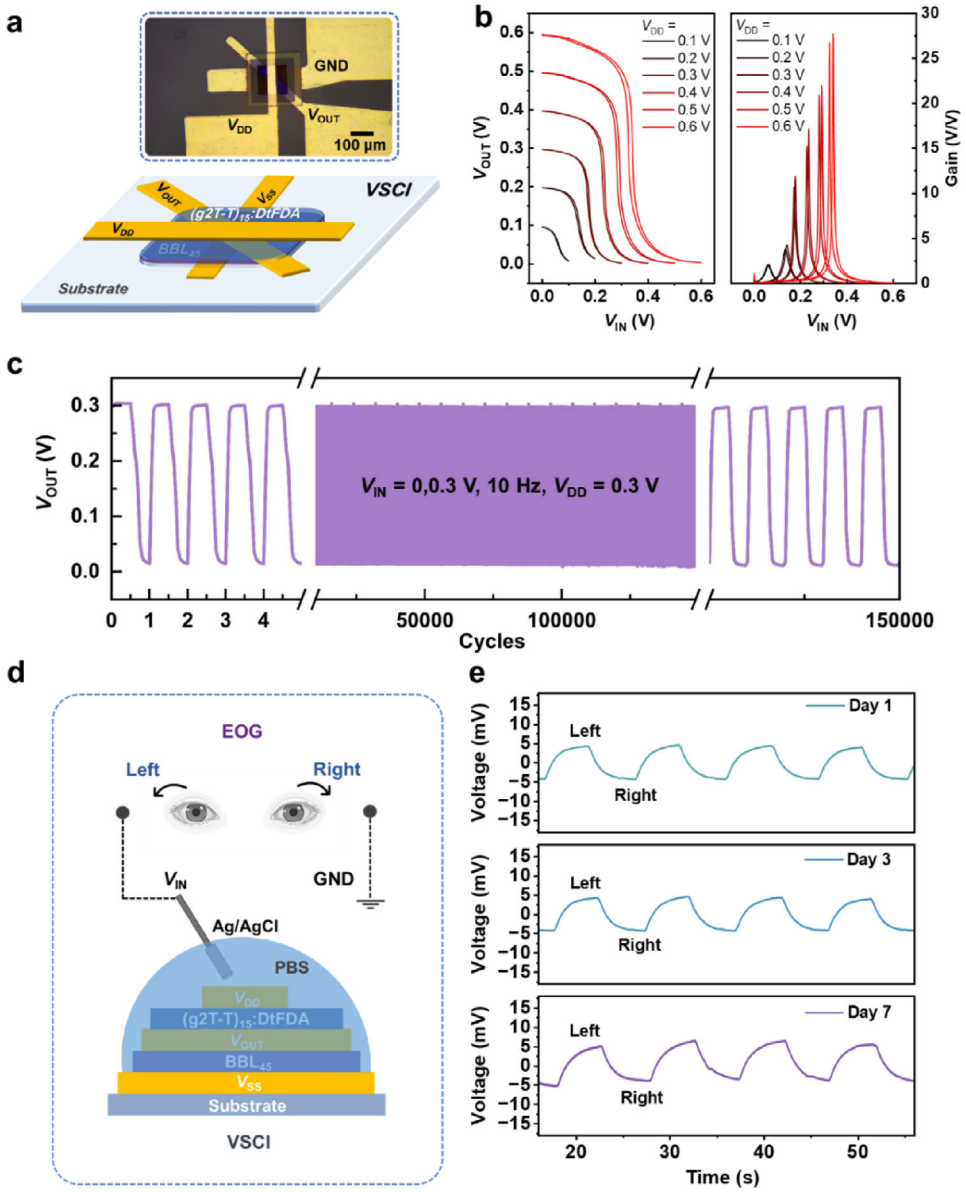


FIGURE 4 | (a) Schematic and optical image of the VSCI. (b) Voltage transfer characteristics (VTC) and voltage gain of the VSCI at various V_{DD} (from 0.1 to 0.6 V). (c) Switching stability of the VSCI for over 150,000 cycles. (d) Schematic diagram of the EOG signals monitoring setup using VSCI. (e) VSCI storage stability characterization by immersing VSCI in 1 × PBS buffer for 7 days and monitoring the EOG signal obtained during left-right eye movements on the first, third, and seventh days.

stannyl)thiophene, tris-(dibenzylideneacetone)-dipalladium (0), tris(o-tolyl)-phosphine, 2-(tributylstannyl)-thiophene, 2-bromothiophene, 1,4,5,8-naphthalenetetracarboxylic dianhydride (NDA), 1,2,4,5-tetraaminobenzene tetrahydrochloride (TABH), poly(phosphoric acid) (PPA), methanesulfonic acid (MSA), anhydrous chlorobenzene, chloroform, n-hexane, methanol, acetone, and isopropyl alcohol were purchased from Sigma–Aldrich. Crosslinker DtFDA was synthesized according to previously reported procedures [39]. BBL with different MWs were synthesized according to a previous report [35]. p(g2T-T) was synthesized following a procedure reported previously [39]. In brief, 5,5′-Dibromo-3,3′-bis(2-(2-(2-methoxyethoxy)ethoxy)ethoxy)-2,2′-bithiophene (250.0 mg, 0.3855 mmol), 2,5-bis(trimethylstannyl)thiophene (158.0 mg, 0.3855 mmol), tris-(dibenzylideneacetone)-dipalladium (0) (17.6 mg, 0.01928 mmol), and tris(o-tolyl)-phosphine (23.5 mg, 0.0771 mmol) were weighed and mixed in a 15 mL sealing tube in a glovebox filled with N_2. Anhydrous chlorobenzene (5 mL) was added via syringe. The sealing tube was then sealed, and the mixture was heated to 200°C and stirred for 6, 10, 16, 20, 24, 30, 36, and 48 h to obtain $(g2T\text{-}T)_{12}$, $(g2T\text{-}T)_{15}$, $(g2T\text{-}T)_{30}$, $(g2T\text{-}T)_{37}$, $(g2T\text{-}T)_{40}$, $(g2T\text{-}T)_{46}$, $(g2T\text{-}T)_{64}$, and $(g2T\text{-}T)_{72}$. Note that, for general Stille coupling polymerization conditions, it is difficult to obtain p(g2T-T) with very high MW due to the formation of an insoluble gel after excessive reaction time. The reaction was then cooled to room temperature, 0.25 mL 2-(tributylstannyl)-thiophene was added, and stirred at 200°C for an additional 0.5 h. Finally, the mixture was again cooled to room temperature, and 0.5 mL 2-bromothiophene was added and stirred at 200°C for 0.5 h before cooling down to room temperature. The deep-colored

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mixture was then dripped into 100 mL of methanol with stirring. After stirring for 30 min, the precipitated solid was collected by filtration and transferred to a Soxhlet thimble. The crude product was subjected to subsequent Soxhlet extractions with n-hexane (3 h), methanol (3 h), acetone (3 h), and chloroform (3 h). After final extraction with chloroform, the polymer solution was concentrated in vacuo to approximately 5 mL and then dripped into 100 mL of methanol with stirring. The polymer was collected by filtration and dried under reduced pressure to afford the product as a dark solid or fibers.

4.2 | Gel Permeation Chromatography

MW distributions of p(g2T-T) were determined using an Agilent PL-GPC 50 system with hexafluoroisopropanol as the eluent.

4.3 | Semiconductor Solution Preparation

All p(g2T-T) batches were dissolved in chloroform with concentrations ranging from 6 to 8 mg mL^{-1}. BBL_{45} and BBL_{105} were dissolved in MSA at 80°C for 12 h with the concentration of 7 and 7 mg mL^{-1}, respectively. All solutions underwent filtration through a polytetrafluoroethylene (PTFE) filter with a diameter of 0.45 μm.

4.4 | Planar OECT Fabrication

Si wafers with 100 nm of SiO_2 as substrates were first treated by ultrasonication in isopropyl alcohol for 15 min, followed by drying with N_2 and treating with UV-ozone (Shenzhen Hwo Technology Co., Ltd) for 15 min. To precisely define the channel geometry of cOECT, the lift-off process was conducted. AZ nLOF 2035 was then deposited on the substrates and photolithographically patterned. Subsequently, 3 nm of Cr and 50 nm of Au were thermally evaporated, then treated with ultrasonic cleaning in isopropyl alcohol for 45 min to pattern the source/drain electrodes. The length and width of the channel are 20 and 500 μm, respectively. The substrates with the electrodes underwent an additional 15-min UV-ozone cleaning. Then, the solution of p(g2T-T) was spin-coated on the substrates at 3000 rpm for 30 s, and the solution of BBL was spin-coated on the substrates at 3000 rpm (acceleration 3000 rpm s^{-1}) for 60 s. Then, the thin BBL films were immersed in deionized water to remove residual MSA and dried with a nitrogen flow. The film thicknesses of p(g2T-T) and BBL with different MWs are carefully controlled to be ∼55 and 35 nm, respectively, by modulating the solution concentrations. After deposition, all semiconductor layers were patterned by a laser direct writing system (R4, LPKF Laser & Electronics AG, Germany) with a laser power of 0.11 W and a frequency of 500 kHz [66]. Finally, the encapsulation layer was then deposited using SU-8 2002 and photolithographically patterned.

4.5 | Vertical OECT Fabrication

vOECTs were also fabricated on pre-cleaned Si/100 nm SiO_2 wafers. First, 3 nm of Cr and 150 nm of Au were thermally evaporated with a shadow mask as the bottom source electrodes. The width of the bottom source electrode is 40 μm. The substrates with the bottom electrodes underwent a 15-min UV-ozone cleaning process before deposition of the semiconductor layer. Then, the solution of p(g2T-T) was spin-coated on the substrates at 3000 rpm for 30 s, and the solution of BBL was spin-coated on the substrates at 3000 rpm (acceleration 3000 rpm s^{-1}) for 60 s. Then, the thin BBL films were immersed in deionized water to remove residual MSA and dried with a nitrogen flow. After deposition, all semiconductor layers were patterned by a laser direct writing system (R4, LPKF Laser & Electronics AG, Germany) with a laser power of 0.11 W and a frequency of 500 kHz. Then, 150 nm of Au was thermally evaporated with a shadow mask as the top drain electrodes while maintaining the substrate at a temperature of approximately 5°C with a back water-cooling system. The width of the top drain electrode is 40 μm. Subsequently, the encapsulation layer was then deposited using SU-8 2002 and photolithographically patterned.

4.6 | Vertically Stacked Complementary Inverter Fabrication

A layer of $(g2T\text{-}T)_{15}$:DtFDA blend (mass ratio of 1:1) was spin-coated directly onto the first BBL_{45}-based vOECT. Note that the width of the bottom and top Au electrodes of the first vOECT are 120 and 40 μm, respectively, and the thicknesses of the bottom and top Au electrodes are both 50 nm. $(g2T\text{-}T)_{15}$:DtFDA layer was then patterned by developing in chloroform after exposure to 365 nm UV-light for 2 min. Next, the third Au electrode (50 nm) was evaporated with a shadow mask as described above. Subsequently, the encapsulation layer was then deposited using SU-8 2002 and photolithographically patterned.

4.7 | OECTs Measurements

Electrical characterization of the OECTs was conducted with an FS-Pro (PDA) semiconductor parameter analyzer along with a probe station (Chengdu Chiptest Technology Co., LTD). A certain volume of PBS solution (about 0.5 μL) was dropped onto the channel region of OECT, and the source and drain electrodes of the devices were connected through probes with an Ag/AgCl pellet applied as the gate electrode. The switching speed of p(g2T-T)-based OECTs was measured by pulsing V_G as a square-wave between +0.3 and −0.5 V and measuring the change in I_D. For cycling stability measurements, a PDMS well was placed on the top of the device's active area, and approximately 50 μL 1 × PBS buffer solution was added and sealed.

4.8 | Human EOG Monitoring

A 28-year-old healthy male participated in the experiment, and all testing procedures were performed safely without any external stimuli. Before the experiment, a human subject was trained on the experimental protocol and was informed not to talk to avoid interference from other body movements. During testing, two disposable electrode patches (3 M 2244) were placed on the left and right sides of the eye. The left electrode was connected to the V_{IN} of the VSCI and the right electrode was grounded. Both V_{DD} and V_{SS} were supplied by an FS-Pro (PDA) semiconductor

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parameter analyzer and properly adjusted to make sure that V_{SW} was ~0 V. Subsequently, in a relaxed state, the subject was asked to gaze horizontally and then was asked to move their eyes to the left for 5 s and then to the right for another 5 s. This procedure was repeated during the measurement. The amplified EOG signal was recorded by monitoring V_{OUT} using the same semiconductor parameter analyzer. For storage stability characterization, the VSCI was washed using deionized water three times and dried with N_2 before EOG monitoring.

4.9 | UV-Vis-NIR Absorption Spectra

The UV-vis-NIR absorption spectra measurements were all conducted on a UNICO UV-4802 spectrophotometer with quartz glass substrates (2 cm × 2 cm) coated with semiconductor thin-films.

4.10 | Spectroelectrochemistry Measurements

The polymer thin films were spin−coated on ITO glass, and the UV-vis-NIR spectra of polymer films were monitored in situ at different voltages (vs. Ag/AgCl) to probe the neutral and charged states during electrochemical doping using a KEITHLEY 2401 SourceMeter. A UV-3600i Shimadzu UV-visible-near-infrared spectrophotometer was utilized, with the light beam passing through the cuvette containing electrolyte aqueous PBS solution (0.01 M) and the polymer/ITO glass sample.

4.11 | Electrochemistry

EIS was performed with a three-electrode setup connected to Gamry Reference 620 electrochemical workstation. Square gold was deposited on a glass substrate as the working electrode (WE), while a platinum mesh served as the counter electrode (CE) and Ag/AgCl as the reference electrode (RE). A polymer film was spin-coated onto the gold substrate and patterned into different areas (from 2.5 cm × 1 cm to 0.625 cm × 1 cm). During EIS testing, the films were fully covered with a 0.01 M PBS solution. Under direct voltage of 0.5 V, EIS spectra were recorded across a frequency range of 100–0.1 Hz with an AC amplitude of 5 mV/rms. The film thickness was determined using AFM to calculate the volume capacitance (C^*). The capacitance of each film was extracted by fitting to the modified Randles circuit model: R_s + CPE / (R_{ct} + Z_w), where R_s is the active electrolyte resistance, CPE is a constant phase element, R_{ct} is the charge transfer resistance, and Z_w is the restricted Warburg diffusion impedance.

4.12 | Grazing-Incidence Wide-Angle X-Ray Scattering

GIWAXS measurements were performed at the BL16B1 beamline of the Shanghai Synchrotron Radiation Facility (SSRF). The X-ray beam with an energy of 10 keV shone upon the samples with the incident angle of 0.15 deg. with respect to the substrate between the critical angles of the organic films and the Si substrate. A custom-made Pilatus-600K detector was placed 167 mm away from the sample to capture GIWAXS images with an exposure time of 30 s.

4.13 | Atomic Force Microscope

AFM images are acquired using a Park NX-7 system in tapping mode under ambient conditions.

4.14 | Inductively Coupled Plasma-Mass Spectrometry

Polymer samples were digested in nitric acid and hydrogen peroxide under elevated temperature (200°C) and pressure in fluoropolymer pressure vessels in a microwave digestion system. After filtration, digested samples were diluted with 2% v/v nitric acid before analysis to target an estimated 1–100 µg/L in the analytical solution. Analysis was performed on a PerkinElmer NexION 2200G ICP-MS.

4.15 | Electrochemical Quartz Microbalance Measurement

EQCM measurements were performed using a gold-coated quartz crystal sensor and a quartz crystal microbalance (QCM922A, Princeton Applied Research). The crystal sensor was first tested without loading in both air and electrolyte conditions (0.01 M PBS). Then, the semiconductor solutions (6–9 mg mL^{-1}) were spin-coated on the crystal surface at 1500 rpm for 1 min. The crystal sensor with coated film was also tested in both air and electrolyte conditions without biasing. Immediately after the stabilization of the polymer-coated sensor in the electrolyte solution, the system was connected to a PRIMARIUS FS Pro source meter for biasing the film. A three-electrode setup comprised an Ag/AgCl RE, a Pt CE, and the polymer-coated sensor as the WE. In the measurements, seven cycles of the −0.3 V → 0 V → +0.5 V→ 0 V bias sequence were applied on the p(g2T-T) films. For relative mass change calculation, the Sauerbrey equation was used to convert $\Delta f/n$ to the change in areal mass (Δm):

$$\frac{\Delta m}{A} = -\Delta f_n \frac{\rho_q V_q}{2 f_0^2 n} \approx \frac{-\Delta f_n}{n} 17.9\ ng\ cm^{-2} \quad (1)$$

where Δf_n is the frequency shift of the nth overtone, A is the sensor active area, ρ_q is the density of quartz, Vq is the shear wave velocity in quartz, f_0 is the fundamental frequency, and n is the overtone number (fixed at 1). Relative mass changes in Figure 3d,e were calculated as the percentage change in total mass relative to the dry mass:

$$\text{Relative mass change} = \frac{m_{total} - m_{dry}}{m_{dry}} \times 100\ \% \quad (2)$$

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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